

INITIAL TEST — 10TH GRADE

Subject: Computer Science — 9th grade review

Student's first and last name: _____ Class: _____

Working time: 50 minutes. 1 point is granted by default. Total score: 10 points.

All subjects are mandatory. Solutions written in C/C++ pseudocode are accepted.

SUBJECT I — Multiple choice items (2 points)

For each item, circle the letter / letters corresponding to the correct variant / variants. Each item can have one, multiple, or all variants correct. (0.5 points each)

1. Which of the following are integer data types in the C/C++ language?

- a) int b) float c) long long d) double

2. Which of the following operators are relational operators in the C/C++ language?

- a) == b) = c) != d) &&

3. Which of the following statements about the for repetitive structure are true?

- a) The for structure is always used when the number of repetitions is not known in advance
b) The instruction `for(i=1;i<=5;i++)` executes the loop body 5 times
c) The increment step of the counter can be negative
d) A while structure cannot be used inside a for structure

4. Which of the following variable declarations are syntactically correct in C++?

- a) `int 2x;` b) `float suma;` c) `char litera = 'a';` d) `int nr_elemente;`

SUBJECT II (3 points)

a) Explain, in natural language, what the adjacent code sequence achieves (what is read, what is calculated/checked, and what is displayed). (1.5 points)

```
int n, ok=1;
cin>>n;
for(int i=2;i<=n/2;i++)
    if(n%i==0) ok=0;
if(n>1 && ok==1)
    cout<<"DA";
else
    cout<<"NU";
```

Answer: _____

b) The adjacent code sequence is given. Complete the trace table (table of values) for the execution of this sequence, for the array $v = \{3, 8, 1, 6, 4\}$. (1.5 points)

```
int v[5] = {3, 8, 1, 6, 4};
int s = 0, c = 0;
for(int i = 0; i < 5; i++)
{
    s = s + v[i];
    if(v[i] > 5)
        c++;
}
```

Complete the table:

i	v[i]	s	c

Final values: $s = \underline{\hspace{2cm}}$ $c = \underline{\hspace{2cm}}$

SUBJECT III — Write complete programs in the C/C++ language (4 points)

1. A natural number n ($n > 0$) is read from the keyboard. Output the number of even digits of n . (2 points)

Example: for $n = 24681$, the displayed result is 4 (the even digits are 2, 4, 6, 8).

2. A natural number n ($1 \leq n \leq 100$) is read from the keyboard, and then n natural numbers. Output, separated by a space, the array elements that are prime numbers. (2 points)

Example: for $n = 5$ and the array 4 7 9 11 6, 7 11 is displayed.